Discovery: What's special about a midsegment?

Triangle #1

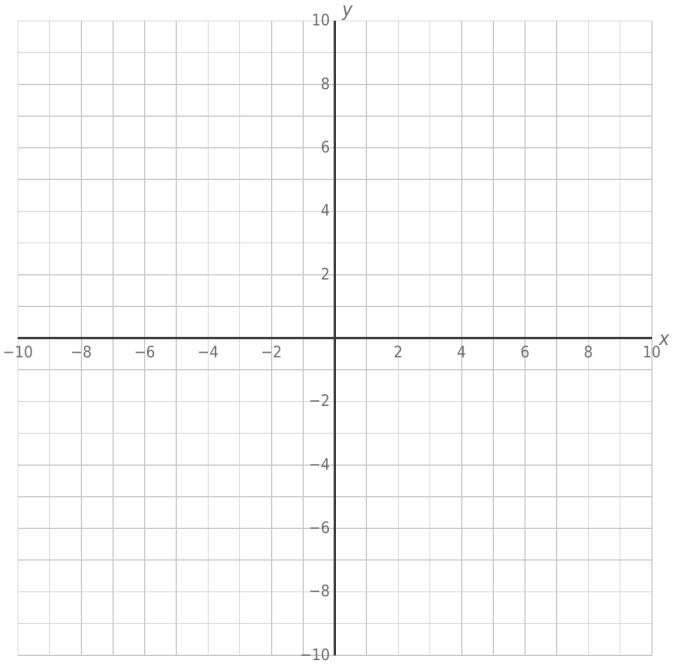
Name: _____ Partner: _____

Vocabulary · fill in the missing word

Midsegment — a segment that connects the _____ of two sides of a triangle

Triangle Midsegment Theorem — after question 5, write it in your own words: _____

Midpoint	Slope	Distance
$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$	$m = \frac{y_2 - y_1}{x_2 - x_1}$	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$



Your triangle

A(2, 10) B(0, 0) C(6, 8)

1. Plot A, B, and C. Connect them.
2. D = midpoint of $\overline{AB}$. E = midpoint of $\overline{AC}$. Plot D and E. Draw segment $\overline{DE}$.

   D = _____ E = _____

3. Find the slopes.

   slope of $\overline{DE}$ = _____

   slope of $\overline{BC}$ = _____

4. Find the lengths.

   DE = _____ BC = _____

Show your work

5. What do you notice about $\overline{DE}$ compared to $\overline{BC}$? Write one complete sentence.

_____

_____

Done early? Find F, the midpoint of $\overline{BC}$. Draw $\overline{DF}$. Compare $\overline{DF}$ to $\overline{AC}$. Does the same pattern hold?

Discovery: What's special about a midsegment?

Triangle #2

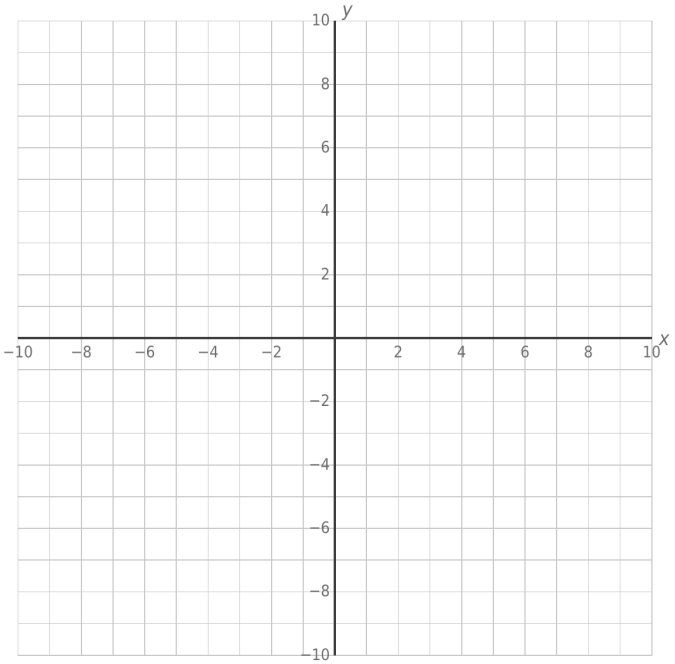
Name: _____ Partner: _____

Vocabulary · fill in the missing word

Midsegment — a segment that connects the _____ of two sides of a triangle

Triangle Midsegment Theorem — after question 5, write it in your own words: _____

Midpoint	Slope	Distance
$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$	$m = \frac{y_2 - y_1}{x_2 - x_1}$	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$



Your triangle

A(−4, 6) B(−6, −2) C(2, −8)

1. Plot A, B, and C. Connect them.
2. D = midpoint of $\overline{AB}$. E = midpoint of $\overline{AC}$. Plot D and E. Draw segment $\overline{DE}$.

   D = _____ E = _____

3. Find the slopes.

   slope of $\overline{DE}$ = _____

   slope of $\overline{BC}$ = _____

4. Find the lengths.

   DE = _____ BC = _____

Show your work

5. What do you notice about $\overline{DE}$ compared to $\overline{BC}$? Write one complete sentence.

_____

_____

Done early? Find F, the midpoint of $\overline{BC}$. Draw $\overline{DF}$. Compare $\overline{DF}$ to $\overline{AC}$. Does the same pattern hold?

Discovery: What's special about a midsegment?

Triangle #3

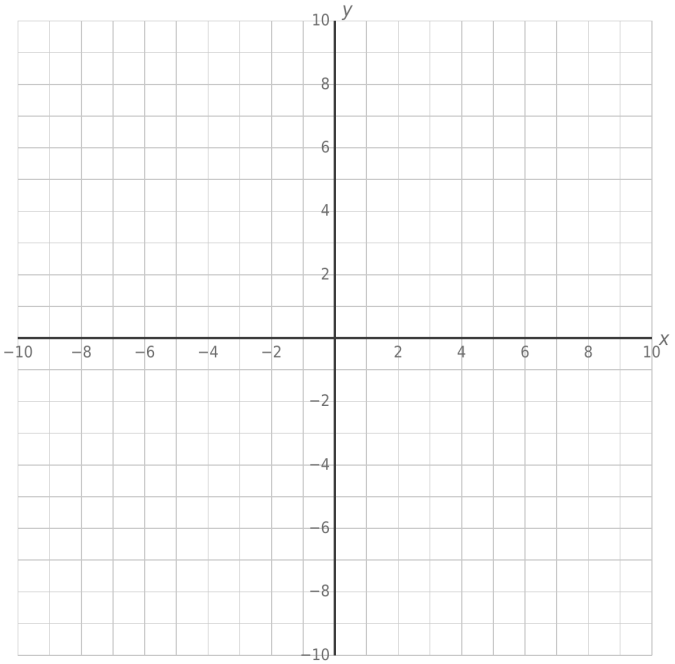
Name: _____ Partner: _____

Vocabulary · fill in the missing word

Midsegment — a segment that connects the _____ of two sides of a triangle

Triangle Midsegment Theorem — after question 5, write it in your own words: _____

Midpoint	Slope	Distance
$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$	$m = \frac{y_2 - y_1}{x_2 - x_1}$	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$



Your triangle

A(4, −2) B(−4, 4) C(4, 10)

1. Plot A, B, and C. Connect them.
2. D = midpoint of $\overline{AB}$. E = midpoint of $\overline{AC}$. Plot D and E. Draw segment $\overline{DE}$.

   D = _____ E = _____

3. Find the slopes.

   slope of $\overline{DE}$ = _____

   slope of $\overline{BC}$ = _____

4. Find the lengths.

   DE = _____ BC = _____

Show your work

5. What do you notice about $\overline{DE}$ compared to $\overline{BC}$? Write one complete sentence.

_____

_____

Done early? Find F, the midpoint of $\overline{BC}$. Draw $\overline{DF}$. Compare $\overline{DF}$ to $\overline{AC}$. Does the same pattern hold?

Discovery: What's special about a midsegment?

Triangle #4

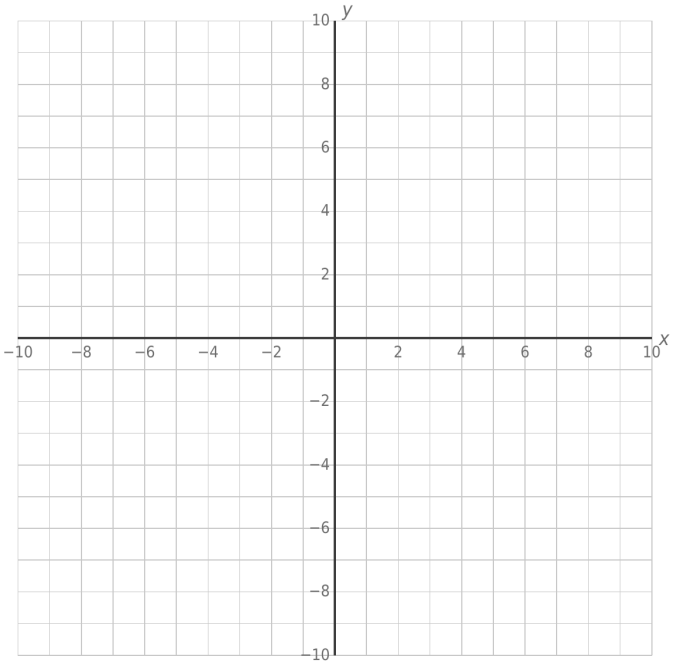
Name: _____ Partner: _____

Vocabulary · fill in the missing word

Midsegment — a segment that connects the _____ of two sides of a triangle

Triangle Midsegment Theorem — after question 5, write it in your own words: _____

Midpoint	Slope	Distance
$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$	$m = \frac{y_2 - y_1}{x_2 - x_1}$	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$



Your triangle

A(−8, 0) B(0, 8) C(6, 0)

1. Plot A, B, and C. Connect them.
2. D = midpoint of $\overline{AB}$. E = midpoint of $\overline{AC}$. Plot D and E. Draw segment $\overline{DE}$.

   D = _____ E = _____

3. Find the slopes.

   slope of $\overline{DE}$ = _____

   slope of $\overline{BC}$ = _____

4. Find the lengths.

   DE = _____ BC = _____

Show your work

5. What do you notice about $\overline{DE}$ compared to $\overline{BC}$? Write one complete sentence.

_____

_____

Done early? Find F, the midpoint of $\overline{BC}$. Draw $\overline{DF}$. Compare $\overline{DF}$ to $\overline{AC}$. Does the same pattern hold?

Discovery: What's special about a midsegment?

Triangle #5

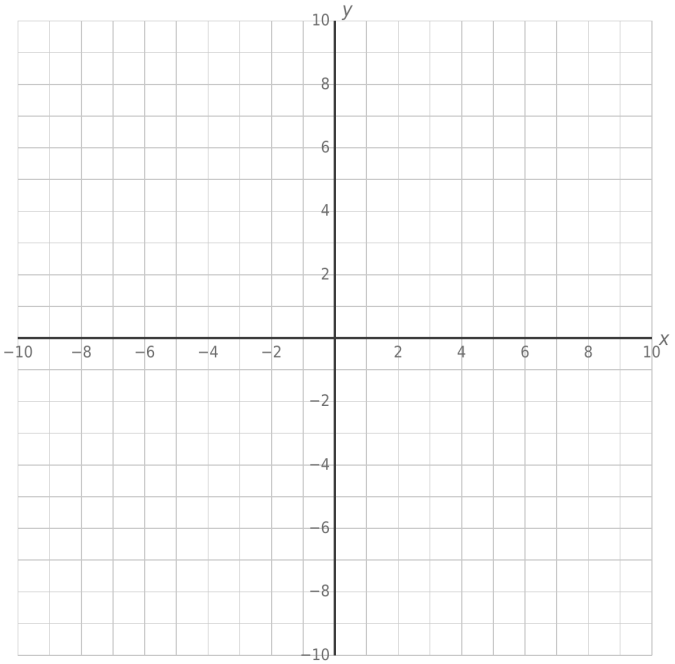
Name: _____ Partner: _____

Vocabulary · fill in the missing word

Midsegment — a segment that connects the _____ of two sides of a triangle

Triangle Midsegment Theorem — after question 5, write it in your own words: _____

Midpoint	Slope	Distance
$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$	$m = \frac{y_2 - y_1}{x_2 - x_1}$	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$



Your triangle

A(0, −6) B(−8, 2) C(2, 2)

1. Plot A, B, and C. Connect them.
2. D = midpoint of $\overline{AB}$. E = midpoint of $\overline{AC}$. Plot D and E. Draw segment $\overline{DE}$.

   D = _____ E = _____

3. Find the slopes.

   slope of $\overline{DE}$ = _____

   slope of $\overline{BC}$ = _____

4. Find the lengths.

   DE = _____ BC = _____

Show your work

5. What do you notice about $\overline{DE}$ compared to $\overline{BC}$? Write one complete sentence.

_____

_____

Done early? Find F, the midpoint of $\overline{BC}$. Draw $\overline{DF}$. Compare $\overline{DF}$ to $\overline{AC}$. Does the same pattern hold?

Discovery: What's special about a midsegment?

Triangle #6

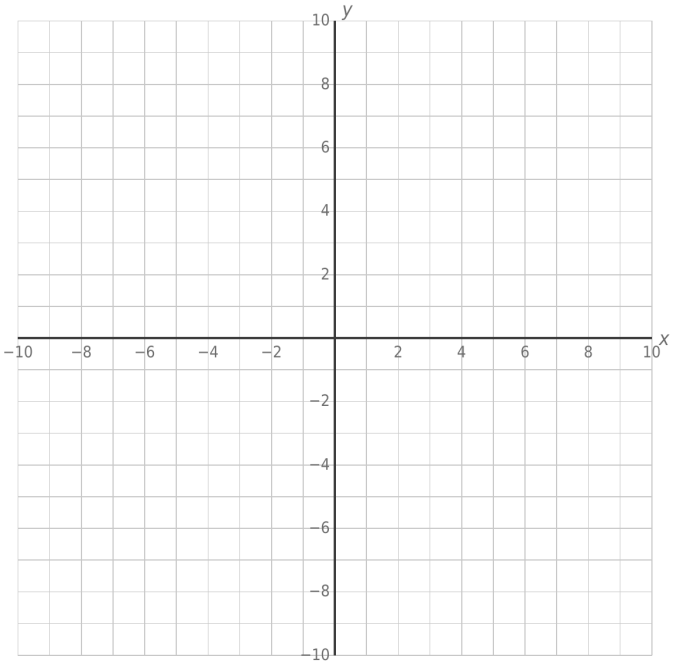
Name: _____ Partner: _____

Vocabulary · fill in the missing word

Midsegment — a segment that connects the _____ of two sides of a triangle

Triangle Midsegment Theorem — after question 5, write it in your own words: _____

Midpoint	Slope	Distance
$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$	$m = \frac{y_2 - y_1}{x_2 - x_1}$	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$



Your triangle

A(10, 6) B(2, −4) C(2, 8)

1. Plot A, B, and C. Connect them.
2. D = midpoint of $\overline{AB}$. E = midpoint of $\overline{AC}$. Plot D and E. Draw segment $\overline{DE}$.

   D = _____ E = _____

3. Find the slopes.

   slope of $\overline{DE}$ = _____

   slope of $\overline{BC}$ = _____

4. Find the lengths.

   DE = _____ BC = _____

Show your work

5. What do you notice about $\overline{DE}$ compared to $\overline{BC}$? Write one complete sentence.

_____

_____

Done early? Find F, the midpoint of $\overline{BC}$. Draw $\overline{DF}$. Compare $\overline{DF}$ to $\overline{AC}$. Does the same pattern hold?